MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 10 (25-29 Oct 2026) - Period 2	Ch 8 - 8.7 Iteration: count-controlled loops

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 10 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Task 8.2: 80% submitted; test tables better. Today: FOR...TO...NEXT loops, STEP, and Python range().

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students write count-controlled loops (FOR ... TO ... STEP ... NEXT) in pseudocode and Python for/range(), and trace them. Syllabus 8.1.4(c); AO2.

Success criteria — *what will successful learning look / sound like?*

I can write a FOR loop with and without STEP; I can match it to Python range() correctly (end value!); I can trace a loop in a trace table.

Key knowledge / skill / understanding:

Count-controlled = fixed number of repeats; range(1, 11) gives 1-10; STEP changes the increment.

Likely misconception / access or challenge need:

range(1,10) includes 10; the loop variable can't be used inside; STEP must be 1.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Brainstorming **Time:** 6 min

What students will do:

List everyday tasks repeated a known number of times (5 daily prayers, 10 push-ups, 30 students' marks). Decide start, end and step for each.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Whiteboards.

Support (low performers / SEN):

Picture prompts.

Stretch (fast learners / gifted):

Find a task that is repeated an unknown number of times.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Code along: times-table printer, countdown with STEP -1, sum of 1..N; then write pseudocode FOR loops and convert to Python.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.7; syllabus pseudocode guide p.45; Python.

Support (low performers / SEN):

Templates with range() hints.

Stretch (fast learners / gifted):

Print a multiplication grid using a loop inside a loop (nested preview).

ACTIVITY 3 *student task + evidence of learning*

Strategy: Problem solving **Time:** 9 min

What students will do:

Trace table challenge: trace a FOR loop with STEP 2 and an accumulator; check with the Python debugger.

Evidence tool: Exit ticket

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Trace table sheet; Thonny debugger.

Support (low performers / SEN):

Partially completed trace table.

Stretch (fast learners / gifted):

Predict the output if STEP is -2.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: write 'I will not talk in class' 100 times - or use a loop. Model FOR Count <- 1 TO 10 ... NEXT Count and Python for count in range(1, 11). Use Thonny's step debugger to watch the counter change.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students watch loop variables change live in the debugger, connecting trace tables to real execution.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Off-by-one with range(); changing the loop variable inside; forgetting NEXT; indentation errors. Likely: 'my loop runs 9 times'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: write a FOR loop that outputs 5, 10, ... 50; give range() for 1 to 20. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.7 count-controlled questions; program: output the sum of the first N odd numbers.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Debugger made loops visible; trace tables improved. Students' everyday examples were rich and personal.

What will I change next time?

Explain range end value with a number line first; most errors were off-by-one.

Students needing follow-up:

10BR1: 2 students absent - send the loop worksheet. Gifted students ready for nested loops.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☑2 ☑3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☑3 ☐C ☐H
Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback